

messaging systems, file systems, and so on are considered services. It allows administrators to create a marketplace of applications and services, which can be used on demand.

Cloud Foundry supports various programming languages like Java, .NET, Ruby, Python, PHP and Go. In Cloud Foundry, the Docker life cycle can be used to deploy Docker images which contain packed applications. The Cloud Foundry Application Runtime can also be deployed to various infrastructure providers like VMware's vSphere, OpenStack, Amazon Web Services, Microsoft Azure, IBM Cloud, Google Cloud Platform and Alibaba Cloud. In addition, some others can also use the runtime using the Cloud Provider Interface (CPI) capability of the Cloud Foundry BOSH project. The Cloud Foundry platform can be used as open source software from the Cloud Foundry Foundation.

The 'Cloud Foundry PaaS Certification programme' from the Cloud Foundry Foundation, which started in December 2015, certifies a Cloud Foundry Certified Provider.

Usually in any PaaS, the physical layer stays at the bottom, and above it are the virtual machines and containers/Kubernetes. The topmost layer is always for the server. However, since Cloud Foundry is serverless, it stays on top of all these layers. This makes it a perfect abstraction for users to develop and deliver applications or services without the need to know about the infrastructure.

Any application or service can be developed using the Cloud Foundry API, and can be deployed and tested in the containers. Such containers are self-maintained by Cloud Foundry's container management systems like Diego and Gardean. In this way, users can easily develop and deliver applications or services to customers without much complexity.

The key features of Cloud Foundry are:

- Application execution, portability, auto-scaling and deployment

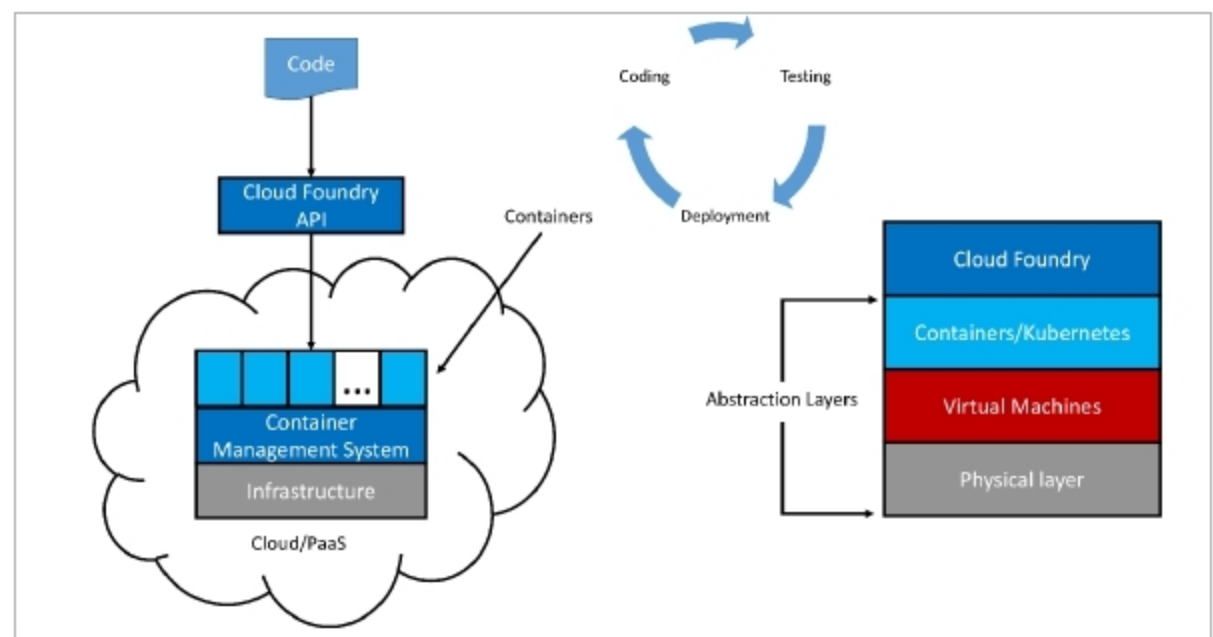


Figure 1: Cloud Foundry Layers

Pros and cons of Cloud Foundry

Pros	Cons
- Easy development, testing and deployment of applications/services using the Cloud Foundry's API. - Supports various programming languages like Java, .NET, Ruby, Python, PHP and Go. - Users can develop applications or services without the need to configure/manage the infrastructure, and Cloud Foundry is an abstraction that eliminates the need to know anything about the infrastructure. - Supports memory and disk space by limiting allocation for individual applications.	- Cloud Foundry does not support stateful containers. - It has logging issues and even though the platform supports showing logs, it doesn't appear to preserve these logs anywhere. - Despite claiming to be a universal solution and supporting various programming languages and frameworks, Cloud Foundry is best suited for applications that are built in accordance with the Twelve-Factor App methodology.

- Application life cycle and health management
- Centralised platform administration and logging
- Provides dynamic routing and external logging of components
- Maintains infrastructure security and cloud provider integration
- Integration with development tools and support for various IaaS providers

Cloud Foundry provides the best solutions and development tools to the user for developing any application or service easily. As this PaaS maintains a high level of abstraction, users do not have to deal much with the underlying infrastructure and can develop applications in various programming languages. **END** 🐧

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